

# Traceable Reproduction Audit and Feedback-Assisted Diagnostics for VLMPC-Style Language-Table Pushing

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**Abstract**—Reliable evaluation of vision-language model predictive control (VLMPC) requires knowing not only whether a task succeeds, but which component selects the target and which component controls the executed action. We introduce a traceable execution-authority audit for a VLMPC-style Language-Table system that separates semantic grounding, planner authority, feedback source, and closed-loop completion across matched runs. The audit reveals a sharp feedback-source bottleneck: raw video-prediction MPC reaches 0/16 at the  $d_{\text{world}}^{\text{final}} \leq 0.08$  tolerance, while a matched semantic controller succeeds in 6/6 runs with oracle state and 0/6 with detector/tracker feedback; the paired visual-minus-oracle final-distance gap is  $0.2548 \pm 0.1453$ . Grounded-controller logs further show that state feedback replaces 23/23 raw actions in every audited run, making execution authority directly observable. The semantic layer is effective when evaluated independently: strict no-cache language trials achieve 9/9 target-selection matches and 9/9 joint controls, and a target-image matrix achieves 8/9 semantic matches and 8/9 joint active successes. Re-query experiments separate mechanism capability from trigger benefit: a forced query repairs 6/6 injected semantic faults and yields 5/6 true-target successes versus 0/6 without re-query, whereas matched natural-event controls preserve identical final distances. Together, these results turn aggregate success rates into component-level evidence and provide a reusable protocol for locating whether a vision-language controller is limited by grounding, action authority, feedback estimation, or recovery logic.

**Keywords**—Vision-language model predictive control, semantic MPC, Language-Table, reproduction audit, event re-query, target-image grounding.

## I. INTRODUCTION

Vision-language manipulation must solve two coupled problems: infer what the instruction refers to and execute how the object should move. Model predictive control (MPC) provides a natural receding-horizon mechanism for the second problem [1], [2], while vision-language models (VLMs) provide semantic grounding for the first [3]–[5]. Their combination is attractive for tasks such as Language-Table pushing, where a controller must identify a named or pictured object, maintain its state, choose a feasible action, and recover when the semantic or visual state becomes stale.

VLMPC couples semantic evaluation with candidate-action prediction inside a visual MPC loop [6]; trajectory-level variants further reduce the burden of full video prediction [7].

Yet an aggregate success rate does not reveal which module produced that success. A correct target can be paired with an ineffective controller; a stable execution can be produced after the raw planner action is replaced; and a re-query can fire without changing the outcome. This attribution problem is especially important in reproduction studies, because semantic correctness, planner authority, feedback quality, and final task completion can otherwise collapse into a single number.

We address this problem with an execution-authority audit that makes four quantities explicit for every branch: the requested and selected target, the source of control state, the action that is actually executed, and the final target-relative outcome. The audit evaluates an unmodified video-prediction chain, a grounded branch that logs and then corrects raw actions, a semantic state-feedback branch, matched oracle and detector/tracker feedback, language and target-image grounding, and event-triggered re-query. Figure 1 summarizes this decomposition. Its central advantage is attribution: each positive or negative result identifies a specific layer rather than being assigned to the full pipeline.

The resulting evidence isolates a clear technical bottleneck. Across seeds 45–50, the same fixed-target semantic controller succeeds in 6/6 runs with oracle state and 0/6 with detector/tracker estimates, with a paired final-distance gap of  $0.2548 \pm 0.1453$ . At the semantic layer, independent strict no-cache trials reach 9/9 language target-selection matches and 8/9 target-image matches. At the recovery layer, forced re-query repairs 6/6 injected semantic faults and reaches 5/6 true-target successes, compared with 0/6 for paired no-requery controls. These results show that the audited system’s strongest separation lies in feedback estimation and execution authority, while semantic grounding and controlled recovery are independently measurable capabilities.

The contributions are:

- 1) An execution-authority protocol that separates semantic selection, state source, raw planner output, executed action, and target-relative success for every controller branch.
- 2) A matched feedback-source experiment that localizes the dominant tested gap: 6/6 success with oracle state

versus 0/6 with detector/tracker estimates under the same semantic controller and seeds.

- 3) Independent grounding and recovery evaluations comprising nine language trials, nine target-image trials, matched natural-event controls, and a six-seed semantic-fault intervention.
- 4) A file-traceable evidence package with per-run CSVs, source and asset hashes, cache accounting, publication figures, and automated consistency checks that bind manuscript claims to executable artifacts.

## II. RELATED WORK

### A. Visual MPC and Video-Prediction Control

Visual MPC uses learned visual dynamics to support model-based manipulation directly in image space. Deep visual foresight and related visual MPC systems showed that action-conditioned video prediction can guide robot motion without a hand-designed dynamics model [8], [9]. Such approaches are attractive for open-ended manipulation, but they are also sensitive to prediction bias, object occlusion, and long-horizon compounding errors. These limitations are relevant to our reproduction: the planner can generate candidate futures and rank actions, but the unmodified closed loop does not reliably drive the object to the selected goal region.

### B. Vision-Language Manipulation

Language-conditioned manipulation has used VLMs and language models for task grounding, affordance selection, reasoning, and failure interpretation. CLIPort separates semantic “what” reasoning from spatial “where” execution [10]. SayCan grounds language plans with affordances [3]. Inner Monologue emphasizes feedback-aware embodied reasoning [5], and VoxPoser composes 3D value maps for robotic manipulation from language-model reasoning [11]. These systems motivate our use of explicit feedback and event handling: semantic decisions must be connected to low-level execution evidence, not only to high-level task labels.

### C. VLMPC, Trajectory Variants, and Testing

VLMPC integrates VLM perception with MPC-style candidate-action evaluation [6], while Traj-VLMPC reduces computational overhead through trajectory generation [7]. Complementary evaluation work such as VLATest emphasizes robustness beyond isolated demonstrations [12], and AHA shows the value of structured failure reasoning [13]. Our work contributes an attribution layer to this evaluation line: it records which module selected the target, which state source informed control, and which action actually reached the environment.

## III. EVALUATION SETUP AND EVIDENCE SOURCES

### A. Task Definition

The experiment uses Language-Table desktop pushing tasks. A target object such as *red moon*, *blue cube*, or *yellow pentagon* should be pushed toward a designated corner. The controller input can be a fixed target label, a natural-language instruction, a target image, or a VLM-based scene selection

request. The action is a planar end-effector push in the simulated table environment.

The primary metric is the final world-coordinate distance between the target object and the goal:

$$\text{success} = \mathbb{I} [d_{\text{world}}^{\text{final}} \leq 0.08] . \quad (1)$$

We use the archived evaluation tolerance of 0.08 for task completion and additionally report 0.05 and 0.10 to expose threshold sensitivity. At 0.05, the grounded suites yield 0/4 and 0/8, while the semantic-MPC suite yields 3/18. The two thresholds therefore distinguish completion under the project criterion from tighter placement precision.

Each live semantic row carries an explicit evaluation scope so that its denominator matches the question being tested. The 18 selected-target control rows comprise 15 fixed-target controls, one natural-language instruction, one target-image interface control, and one scene-selection control. Language grounding is evaluated on the instruction row (1/1 requested-to-selected match and joint success), while target-image grounding is evaluated on the independent nine-run matrix (8/9); the aggregate 18/18 reports selected-target control completion.

For the new target-image matrix, we distinguish a *pre-satisfied* run, for which the true target is already within the threshold before the first action, from an *active control* trial, which starts outside the threshold and executes at least one action. *Joint active success* requires both correct target-image semantic selection and active control success. This prevents a zero-step terminal state from being counted as controller evidence.

### B. Evidence Artifacts

All numerical results in this paper are extracted from the study package. The main evidence files are:

- sanitized per-run and aggregate CSVs for the fresh raw-MPC sweep, semantic suite, target-image matrix, event controls, semantic-fault audit, and visual-feedback test;
- supplement CSVs for exact method parameters, checkpoint/code hashes, target-image selection, and VLM cache status;
- publication figures and selected qualitative frames used as visual evidence; raw service logs and videos remain outside the sanitized public package;
- a portable robustness-edge audit CSV covering occlusion, viewpoint proxy, brightness shift, and unknown-target rejection, together with its four input images.

These artifacts bind every reported quantity to a run-level record. Formal VLM calls use the project’s CodexCLIClient with the codex backend and GPT-5.5 alias; an isolated CODEX\_HOME supplies runtime configuration, and strict no-cache controls record zero cache hits. The public v1.0.10 evidence package at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit> includes portable inputs, reproduction instructions, parameters, figure sources, and separate runtime-asset and source-snapshot

manifests. Third-party model weights are obtained from their original sources.

#### IV. METHOD

##### A. Execution-Authority Audit

We represent each run by the tuple  $(g_{\text{req}}, g_{\text{sel}}, s_t, a_t^{\text{raw}}, a_t^{\text{exec}}, y)$ : the requested target, selected target, control-state source, raw planner action, executed action, and terminal outcome. This representation separates three questions that an aggregate success rate conflates: whether semantic grounding selects the intended object, whether the planner or a feedback law controls the environment, and whether the resulting trajectory reaches the target. Figure 1 maps the common input and logging pipeline to the evidence branches.

The branches act as attribution probes. The raw branch executes  $a_t^{\text{raw}}$ ; the grounded branch logs that action but executes a state-feedback correction; the semantic branch combines target selection with a state-feedback controller; and the visual-feedback branch replaces oracle state with detector/tracker estimates while retaining the semantic controller. Matched seeds and explicit state-source labels make the transition from semantic correctness to closed-loop execution directly measurable.

##### B. Unmodified Reproduction Chain

The unmodified reproduction follows a VLMPC-style loop. A task input is grounded to a target, candidate action sequences are sampled, a visual prediction model forecasts future observations, and a cost function ranks the candidates before the first action is executed. Controller comparison includes baseline, improved, literature, and literature-v2 settings. Parameter ablation varies sampling budget and cost settings.

This branch applies no final-action state-feedback correction, so the raw candidate-ranking output retains execution authority. It provides the reference point for the feedback-source and action-authority comparisons that follow.

##### C. Codex CLI VLM Adapter

Formal online VLM calls are executed through the local CodexCLIClient. For each `complete(prompt, image_paths)` call, the adapter stages image attachments in an ASCII-only temporary directory when the project path contains non-ASCII characters, sends the prompt through standard input, and launches an ephemeral CLI request equivalent to

```
codex exec -skip-git-repo-check -ephemeral
           -color never
-C <ASCII workdir> -o <temporary output>
-m gpt-5.5 -i <image>
```

The response is read from the temporary output file and parsed for target, subtask, or re-query state. An isolated CODEX\_HOME supplies runtime configuration while keeping credentials outside the source tree. When enabled, the cache

TABLE I  
RECOVERABLE IMPLEMENTATION PARAMETERS FROM CODE, METRICS,  
AND SUPPLEMENT CSVS.

| Parameter         | Value or rule used in the combined audit                                                                                                                         |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environment       | BLOCK_8 Language-Table, 10 Hz; fresh raw sweep uses seeds 45–48; semantic, target-image, and event controls use 42–44; visual feedback uses 45–50                |
| Raw MPC           | action dim. 2, horizon 20; sampled candidates clipped to $[-0.02, 0.02]$ , executed action/feedback bound $[-0.08, 0.08]$                                        |
| Budgets           | default 200/5 samples/plan frequency; eight archived ablations: pf1/ns5, pf2/ns5, pf2/ns10, pf3/ns10, pf2/ns15, cost history low, cost default, cost target high |
| Detector/tracker  | YOLO conf. 0.5; PySOT SiamRPN tracker                                                                                                                            |
| Cost weights      | contact/approach/alignment/obstacle = 0.35/0.15/18.0/24.0                                                                                                        |
| Progress terms    | reward 1.2, non-progress penalty 12.0, min gain 1.0, obstacle decay 42 px                                                                                        |
| Close range       | distance 135 px; contact/alignment/reward/penalty = 0.10/6.0/1.8/20.0                                                                                            |
| Feedback push     | contact offset 0.06, clearance 0.12, move step 0.06, push step 0.075, tolerance 0.018                                                                            |
| Original event    | default window/min-progress/min-step = 3/1.0/3; semantic matched controls use 4/0.002/3 with min progress in world-distance units                                |
| Semantic event    | window 4, min progress 0.002 world-distance units, distance floor $1.5 \times$ success distance                                                                  |
| Online VLM        | CodexCLIClient; local codex exec -m gpt-5.5; isolated CODEX_HOME; credentials excluded                                                                           |
| Success threshold | evaluation tolerance $d_{\text{world}} \leq 0.08$ ; strict audit also reports 0.05 and 0.10                                                                      |

key combines the model configuration, prompt, and image-content hashes; all formal event and language controls disable the cache and record zero cache hits. The CLI path supplies the evidence reported here, while an OpenAI-compatible backend remains available for development.

##### D. Archived Implementation Parameters

Table I summarizes the evaluated implementation parameters. The unmodified planner uses horizon  $H = 20$ , action dimension 2, and a correlated-noise sampler with  $\beta = 0.5$ . The command-line defaults are 200 samples and replan frequency 5, and the archived ablation contributes eight recorded sampling and cost configurations. The literature-style cost is

$$J = d_{\text{obj,goal}} + 0.35d_{\text{contact}} + 0.15d_{\text{arm,obj}} + 18.0J_{\text{align}} + 24.0J_{\text{obs}} - 1.2\Delta d + 12.0\mathbb{I}[\Delta d < 1.0], \quad (2)$$

with obstacle decay 42 pixels. In the close-range branch, the contact, alignment, progress-reward, and non-progress terms are changed to 0.10, 6.0, 1.8, and 20.0, respectively. These recovered values define the evaluated implementation.

The runtime-asset manifest records SHA-256 prefixes for the externally obtained video-prediction, detector, and tracker assets. The source manifest separately records the executed-source and public-snapshot hashes. Five listed Python files are byte-identical. For `main.py`, the executed-source prefix is `d8078e7c8a86` and the public-snapshot prefix is `f98caece7442`; the difference is limited to replacing a drive-local dependency fallback with environment-based configuration, without changing controller logic.

##### E. Grounded Original MPC

Grounded original MPC isolates the effect of final-action feedback while preserving the raw planner trace. Each raw

MPC action is recorded before a bounded simulator-state correction determines the action executed in the environment. Across the 12 grounded runs (4 controller-comparison runs and 8 ablation runs), every run contains 23 raw planner actions and 23 feedback overrides. The 23/23 replacement rate makes the transfer of execution authority directly observable instead of attributing the resulting trajectory to the raw planner.

The feedback action uses object, end-effector, and goal positions from simulator state. Let  $\hat{d}_{\text{push}} = (x_{\text{goal}} - x_{\text{obj}}) / \|x_{\text{goal}} - x_{\text{obj}}\|$  and  $x_{\text{contact}} = x_{\text{obj}} - 0.06\hat{d}_{\text{push}}$ . The controller first stages the end effector near a side-clearance point, then moves to contact, and finally pushes:

$$u_t = \text{clip} \left( \alpha \hat{d}_{\text{push}} + \beta (x_{\text{contact}} - x_{\text{ee}}), -0.08, 0.08 \right). \quad (3)$$

In the push phase,  $\alpha = 0.075$  and  $\beta = 0.45$ ; the stage/contact phases instead use a step-limited move of at most 0.06.

#### F. Semantic MPC

Semantic MPC isolates target selection from low-level state-feedback execution. It accepts fixed labels, language instructions, target images, or free VLM scene selection, then converts the selected target into a bounded feedback push. The `auto|vlm|vision` target-image interface sends the target image and current scene image to the configured VLM and asks it to choose among visible candidates. The branch retains receding-horizon execution and logging, but activates no raw video-prediction sampling budget ( $N_{\text{raw}} = 0$ ); the state-feedback law has execution authority. This separation tests whether a selected semantic target can be converted into stable closed-loop control when state is available.

#### G. Event-Triggered VLM Re-query

Event-triggered re-query tests when semantic state should be refreshed. For the semantic branch, progress is measured in world-coordinate target distance using a stagnation window of 4 and a minimum progress threshold of 0.002. Fresh matched controls compare no-event and event-enabled runs without forcing the initial query. All controls disable the cache, record zero cache hits, and log fresh `CodexCLIClient` responses. Natural re-query fires in all three event-enabled runs; seed-matched final distances remain identical to the no-event controls. This comparison separates correct trigger execution from incremental benefit under conditions that already succeed.

A paired fault intervention then measures recovery capacity. Each of six seeds begins with the true target replaced by a deliberately wrong target (blue cube instead of red moon). The treatment issues one explicitly labeled audit re-query at step 3, whereas the control retains the corrupted target. Evaluation uses the independently specified true target. The `CodexCLIClient` treatment repairs the semantic state in 6/6 runs and reaches the true-target threshold in 5/6, compared with 0/6 for the no-requery controls. The intervention therefore provides direct evidence that re-query can restore corrupted semantic state.

An independent language-repeat audit separates language grounding from aggregate controller success. Three instructions (red moon, blue cube, and yellow pentagon)

are each evaluated at seeds 42–44 through the same `CodexCLIClient` path, with the expected target specified independently of the controller output. All nine strict no-cache runs produce one fresh VLM response, match the expected target, and meet the 0.08 control threshold.

## V. EXPERIMENTS

### A. Evidence Design

Three evidence groups support distinct attribution questions. **Raw planning** evaluates the four-variant, four-seed video-prediction chain and eight recorded parameter configurations. **Execution and semantic control** measures action replacement, language and image grounding, and recovery under oracle state feedback. **Feedback source** holds the semantic controller and seeds fixed while replacing oracle state with detector/tracker estimates. This design aligns each success rate with the component whose behavior it measures.

### B. Raw-Chain Reference

Table III establishes the raw-chain reference. Across four controller variants and seeds 45–48, all 16 runs remain above the 0.08 tolerance, with variant means from 0.3134 to 0.3358 and an aggregate mean of 0.3245. The outcome is unchanged at 0.05 and 0.10, and the eight recorded parameter configurations also yield 0/8. In combination with the matched feedback-source experiment below, these results rule out the tested controller and parameter variants as sufficient explanations for stable execution.

### C. Execution-Authority and Feedback-Source Attribution

Table IV identifies the subsystem that changes the outcome. Under the same semantic controller and seeds 45–50, oracle feedback reaches 6/6 while detector/tracker feedback reaches 0/6; the paired visual-minus-oracle final-distance gap is  $0.2548 \pm 0.1453$ . Grounded logs provide the complementary authority trace: all 12 runs replace 23/23 raw planner actions before execution. The broader semantic suite reaches 18/18 selected-target controls at the 0.08 criterion, with 15/15 fixed-target controls, one language control, one target-image interface control, and one scene-selection control. The independent target-image matrix contributes 8/9 semantic matches and joint active successes, while the language repeat audit contributes 9/9 matches and 9/9 joint controls.

The branch comparison therefore separates three claims: semantic target selection can be evaluated independently; oracle state feedback can stabilize execution; and replacing oracle state with the implemented visual estimates is the dominant tested bottleneck. At 0.05, the selected-target semantic suite reaches 3/18, so the 0.08 and 0.05 columns jointly report completion and placement precision.

### D. Claim-to-Run Evidence

Table VI links each aggregate claim to portable run records containing seeds, labels, final distance, steps, runtime, request counts, selector metadata, and sanitized provenance. The digest makes the contrast across raw planning, oracle

TABLE II  
CONTROLLER BRANCHES USED TO ATTRIBUTE SEMANTIC ROLE, STATE SOURCE, AND EXECUTED-ACTION AUTHORITY.

| Branch                   | Semantic/planning role                                                            | Actual MPC role                                                                                      | Executed action authority                                                           | State source             |
|--------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| Unmodified VLMPC         | Ground target; sample action sequences; predict visual futures                    | Candidate ranking with a video-prediction cost; first raw action is executed                         | Raw planner action                                                                  | Visual observations only |
| Grounded original MPC    | Preserve the same raw planner call and log its actions                            | Raw planner remains in the loop for comparison and traceability                                      | Privileged state-feedback correction replaces the raw action (23/23 in all 12 runs) | Simulator state          |
| Semantic MPC             | Select target from label, language, image, or VLM scene query                     | Receding-horizon execution/logging; no active raw video-prediction sampling ( $N_{\text{raw}} = 0$ ) | Privileged state-feedback push law                                                  | Simulator state          |
| Visual feedback          | Select a fixed target; estimate control state from RGB observations               | Receding-horizon detector/tracker feedback; no simulator-state accessor                              | Visual detector/tracker feedback law                                                | RGB observations         |
| Event-control supplement | Monitor stagnation/semantic events and refresh the VLM instruction when triggered | Event logic controls query timing, not the low-level push optimizer                                  | Same semantic-MPC feedback action                                                   | Simulator state          |

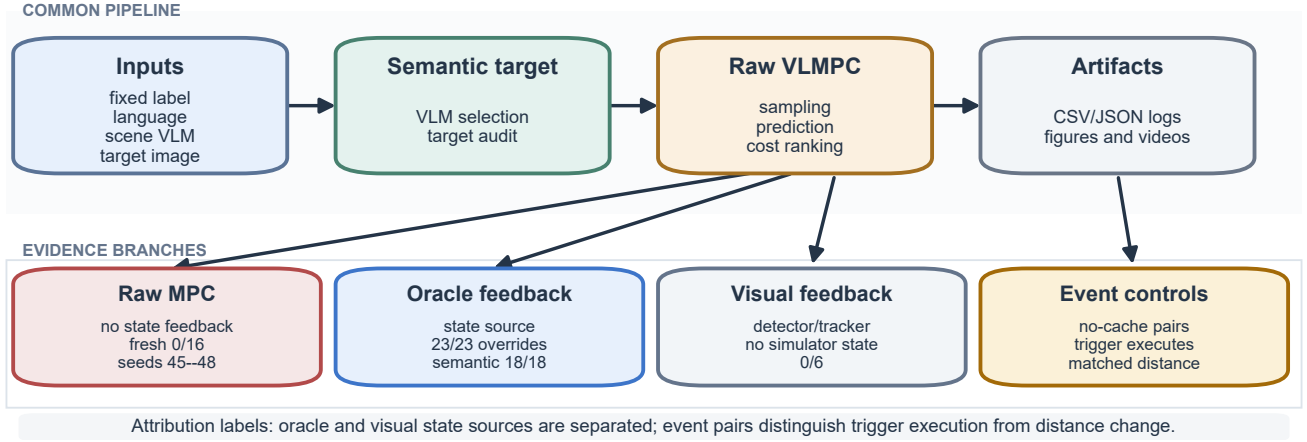


Fig. 1. Execution-authority overview. Explicit state-source and action-authority labels separate raw planning, semantic selection, feedback execution, and re-query recovery. Matched event controls distinguish trigger execution from recovery benefit.

TABLE III  
RAW-CHAIN REFERENCE ACROSS SEEDS 45–48 AND EIGHT RECORDED PARAMETER CONFIGURATIONS. FINAL DISTANCE IS MEAN  $\pm$  SAMPLE SD; THE FRESH SWEEP AND ARCHIVED CONFIGURATIONS RETAIN SEPARATE DENOMINATORS.

| Variant/suite      | Runs | Succ. | Final dist.         | Range       |
|--------------------|------|-------|---------------------|-------------|
| Baseline           | 4    | 0/4   | $0.3358 \pm 0.1099$ | 0.199–0.436 |
| Improved           | 4    | 0/4   | $0.3354 \pm 0.1048$ | 0.224–0.436 |
| Literature         | 4    | 0/4   | $0.3134 \pm 0.1325$ | 0.163–0.436 |
| Literature-v2      | 4    | 0/4   | $0.3134 \pm 0.1325$ | 0.163–0.436 |
| All fresh raw runs | 16   | 0/16  | $0.3245 \pm 0.1085$ | 0.163–0.436 |
| Archived ablation  | 8    | 0/8   | $0.2534 \pm 0.0304$ | 0.216–0.294 |

Threshold sensitivity at 0.05/0.08/0.10 is 0/16, 0/16, 0/16 for the fresh raw sweep and 0/8, 0/8, 0/8 for the archived ablation.

execution, visual feedback, grounding, and recovery visible without merging their denominators.

#### E. Re-query Trigger and Recovery

The event study separates trigger behavior from recovery. The 2026-09-22 set contains three matched no-event controls, three event-enabled controls, and three forced scene-selection audits at seeds 42–44. Natural re-query fires in 3/3 enabled runs, while no-event and event-enabled runs retain identical seed-matched final distances. Every row uses strict no-cache mode and records zero cache hits. The paired semantic-fault intervention then supplies the missing recovery condition by

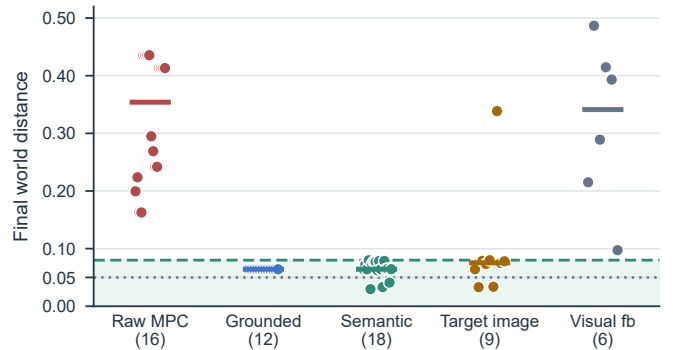


Fig. 2. Final-distance distributions across the fresh raw-MPC sweep, archived grounded diagnostics, live semantic and target-image suites, and the visual-feedback test. Each dot is one run and the horizontal segment is the within-suite median.

evaluating both treatments against an independently specified true target.

Under the injected fault, re-query restores the correct target in 6/6 runs and reaches the true-target threshold in 5/6, whereas the paired controls remain at 0/6. One treatment run ends just above 0.08. Together, the natural controls show that the trigger path executes, while the intervention shows that a fresh

TABLE IV

BRANCH-LEVEL RESULTS FOR EXECUTION AUTHORITY, SEMANTIC GROUNDING, AND FEEDBACK SOURCE. STATE SOURCE IS REPORTED EXPLICITLY SO MATCHED ORACLE AND VISUAL ROWS CAN BE COMPARED WITHOUT POOLING DISTINCT CONTROLLERS.

| Suite                    | Runs      | Control succ. | Evidence and semantic check                     | Final dist.                   | State source |
|--------------------------|-----------|---------------|-------------------------------------------------|-------------------------------|--------------|
| Grounded controller      | 4         | 4/4           | target fixed; 23/23 override                    | $0.0643 \pm 0.0000$           | oracle       |
| Grounded ablation        | 8         | 8/8           | target fixed; 23/23 override                    | $0.0643 \pm 0.0000$           | oracle       |
| Event controls           | 9         | 9/9           | matched controls; natural trigger 3/3 enabled   | $0.0587 \pm 0.0233$ matched   | oracle       |
| Semantic MPC live        | 18        | 18/18*        | 15/15 fixed; language 1/1; image 1/1; scene 1/1 | $0.0641 \pm 0.0150$           | oracle       |
| Target-image matrix      | 9         | 8/9 joint     | semantic 8/9; no pre-satisfied                  | $0.0950 \pm 0.0932$           | oracle       |
| Oracle feedback (paired) | 6         | 6/6           | fixed target; seeds 45–50                       | $0.0612 \pm 0.0206$           | oracle       |
| Visual feedback          | 6         | 0/6           | same semantic controller; seeds 45–50           | $0.3160 \pm 0.1439$           | visual       |
| Semantic-fault audit     | 12 paired | 5/6 treatment | repair 6/6; control 0/6                         | $0.0639 \pm 0.0157$ treatment | oracle       |
| Language repeats         | 9         | 9/9 joint     | 3 instructions x 3 seeds; Codex no-cache        | $0.0661 \pm 0.0191$           | oracle       |

\* The 18-row semantic suite measures selected-target control: 15 fixed-target controls and one each for language, target-image interface, and free scene selection. Independent grounding rates are reported by the nine-run language and target-image evaluations.

TABLE V

CONDITION-WISE VARIABILITY FOR THE SEMANTIC-MPC SUITE. VALUES ARE MEAN  $\pm$  SAMPLE SD OF FINAL WORLD DISTANCE; GROUPS IDENTIFY INPUT-FORM, TARGET, SEED, AND CONTROLLER-SETTING COVERAGE.

| Condition         | $n$ | Mean $\pm$ SD       | $\leq 0.05$ | $\leq 0.08$ | Target/seed scope        | Interpretation                    |
|-------------------|-----|---------------------|-------------|-------------|--------------------------|-----------------------------------|
| Input forms       | 4   | $0.0663 \pm 0.0041$ | 0/4         | 4/4         | 3 forms + target image   | Input-form comparison             |
| Multi-target      | 8   | $0.0678 \pm 0.0167$ | 1/8         | 8/8         | 8 target objects         | Small target sweep                |
| Multi-seed        | 3   | $0.0587 \pm 0.0233$ | 1/3         | 3/3         | red moon, seeds 42/43/44 | Seed variability                  |
| Semantic ablation | 3   | $0.0568 \pm 0.0136$ | 1/3         | 3/3         | default/precise/fast     | Semantic-control settings         |
| All latest runs   | 18  | $0.0641 \pm 0.0150$ | 3/18        | 18/18       | heterogeneous union      | Aggregate selected-target control |

TABLE VI

PER-RUN DIGEST EXTRACTED FROM ARCHIVED AND LATEST CSV FILES. EXEC. S IS DISTANCE-THRESHOLD SUCCESS UNDER  $d_{\text{world}}^{\text{final}} \leq 0.08$ .

| Suite                         | Final-distance summary                                          | Range        | Exec. S       | Semantic/VLM evidence                                    |
|-------------------------------|-----------------------------------------------------------------|--------------|---------------|----------------------------------------------------------|
| Fresh raw MPC                 | 4 variants $\times$ 4 seeds; aggregate mean $0.3245 \pm 0.1085$ | 0.163–0.436  | 0/16          | raw visual-prediction chain                              |
| Archived raw ablation         | eight logged parameter settings                                 | 0.216–0.294  | 0/8           | archived comparison                                      |
| Grounded controller           | all 0.064                                                       | 0.064–0.064  | 4/4           | fixed target                                             |
| Grounded ablation             | all 0.064                                                       | 0.064–0.064  | 8/8           | fixed target                                             |
| Event controls                | matched no-event/event distances identical by seed              | 0.033–0.079  | 9/9           | natural trigger 3/3; zero incremental distance change    |
| Semantic MPC live             | min 0.030; median 0.064; max 0.0798                             | 0.030–0.0798 | 18/18         | 15/15 fixed controls; language 1/1; image 1/1; scene 1/1 |
| Target-image matrix           | 8 semantic matches; one yellow-pentagon error                   | 0.033–0.339  | 8/9 joint     | all nine are active trials                               |
| Visual feedback               | detector/tracker feedback, seeds 45–50                          | 0.097–0.487  | 0/6           | no simulator-state access                                |
| Semantic-fault revision audit | treatment repairs target in 6/6; control 0/6                    | 0.044–0.385  | 5/6 treatment | synthetic re-query; natural count 0                      |
| Language repeats              | three instructions x seeds 42–44; 9/9 joint                     | 0.034–0.0799 | 9/9           | Codex no-cache; independent labels                       |

TABLE VII

FRESH EVENT RE-QUERY CONTROLS FROM THE STRICT NO-CACHE 2026-09-22 EXPERIMENT.

| Case                                | Natural query    | re- query | Final dist.   | Note                            |
|-------------------------------------|------------------|-----------|---------------|---------------------------------|
| No-event controls, seeds 42–44      | 0                |           | 0.0332–0.0787 | 3/3 success; 3 fresh responses  |
| Event-enabled controls, seeds 42–44 | 1 each           |           | 0.0332–0.0787 | 3/3 success; 6 fresh responses  |
| Forced scene audit, seeds 42–44     | natural in all 3 |           | 0.0339–0.0725 | 3/3 success; 14 fresh responses |

query can recover a corrupted semantic state when recovery is required.

### F. Target-Image Grounding

The VLM selector improves target-image grounding from 1/4 matches with the detector/corner heuristic to 4/4 in the

TABLE VIII

CONTROLLED SEMANTIC-FAULT RE-QUERY AUDIT. THE WRONG INITIAL TARGET IS INJECTED DELIBERATELY; SUCCESS IS MEASURED AGAINST THE INDEPENDENT TRUE TARGET.

| Condition               | $n$ | Semantic repair | True-target success | Final dist.         |
|-------------------------|-----|-----------------|---------------------|---------------------|
| No re-query control     | 6   | 0/6             | 0/6                 | $0.2435 \pm 0.0904$ |
| Forced re-query (Codex) | 6   | 6/6             | 5/6                 | $0.0639 \pm 0.0157$ |

four-case semantic audit. It receives the target image and current scene image, selects among visible scene objects, and records the raw response. The end-to-end matrix then repeats three targets across seeds 42–44, yielding 8/9 semantic matches and 8/9 joint active successes. The retained yellow-pentagon error identifies the remaining grounding failure surface.

| Threshold sensitivity       |             |             |             |
|-----------------------------|-------------|-------------|-------------|
|                             | $\leq 0.05$ | $\leq 0.08$ | $\leq 0.10$ |
| Fresh raw MPC (n=16)        | 0/16        | 0/16        | 0/16        |
| Archived ablation (n=8)     | 0/8         | 0/8         | 0/8         |
| Grounded diagnostics (n=12) | 0/12        | 12/12       | 12/12       |
| Semantic MPC (n=18)         | 3/18        | 18/18       | 18/18       |
| Target image (n=9)          | 2/9         | 8/9         | 8/9         |
| Visual feedback (n=6)       | 0/6         | 0/6         | 1/6         |

Fig. 3. Threshold sensitivity at 0.05, 0.08, and 0.10. The outlined 0.08 column is the project completion criterion: raw MPC remains 0/16, whereas selected-target semantic control reaches 18/18. The stricter 0.05 column (3/18) exposes the remaining placement-precision gap.

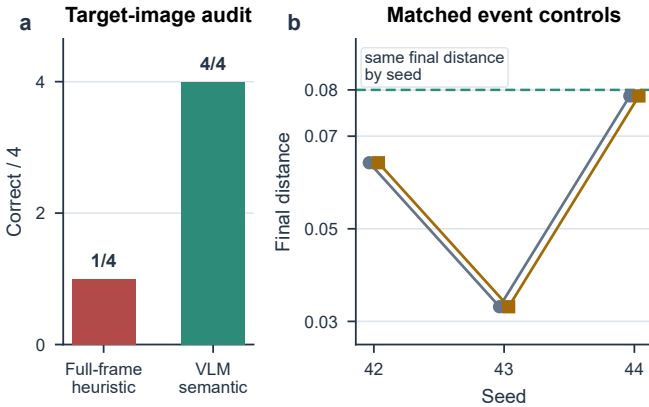


Fig. 4. Target-image and event-control audits. (a) The live VLM semantic selector is correct in all four audited cases, whereas the old full-frame heuristic is correct in only one. (b) Fresh strict no-cache matched controls have identical no-event and event-enabled final distances by seed; all nine rows record zero cache hits.

TABLE IX  
TARGET-IMAGE SEMANTIC-SELECTION AUDIT. VLM SEMANTIC CHOICE USES THE PROJECT VLM BACKEND WITH TARGET-IMAGE AND SCENE-IMAGE INPUTS.

| Case              | Exp.            | Old       | VLM             | OK  |
|-------------------|-----------------|-----------|-----------------|-----|
| Red-moon crop     | red moon        | blue cube | red moon        | yes |
| Blue-cube crop    | blue cube       | blue cube | blue cube       | yes |
| Yellow-pent. crop | yellow pentagon | blue cube | yellow pentagon | yes |
| Archived frame    | red moon        | blue cube | red moon        | yes |

### G. Target-Image End-to-End Evaluation

The end-to-end evaluation uses three manually specified target crops and seeds 42–44. The VLM selects a named scene object from the target and scene images, after which the semantic-MPC controller executes the selected target. All nine trials begin outside the 0.08 threshold and require active control. Eight select and control the intended object successfully; the yellow-pentagon run at seed 42 selects *yellow star* and ends at 0.3386.

The complete per-run table is included in the target-image

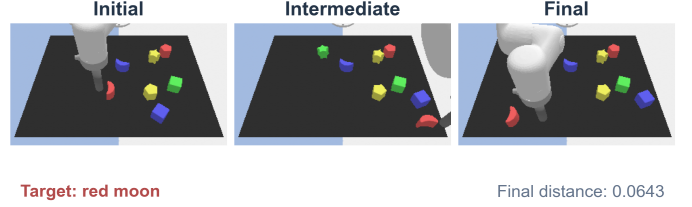


Fig. 5. Representative frames from a logged oracle-feedback red-moon execution; quantitative outcomes are reported in the portable run tables.

CSV. The matrix expands the interface evaluation beyond a single run, and the retained miss localizes its error to semantic selection rather than low-level execution.

### H. Bottleneck Attribution

The factor-isolating comparisons in Table XI localize the dominant tested bottleneck. Controller and parameter variants leave the raw chain above tolerance, whereas direct action correction succeeds with a 23/23 replacement trace. Holding the semantic controller and seeds fixed then produces the decisive contrast: 6/6 with oracle state and 0/6 with visual estimates. Semantic grounding is comparatively strong but not saturated, as shown by 9/9 language matches and 8/9 target-image matches.

### I. Perception Stress Tests

The perception stress test probes target occlusion, a viewpoint proxy, brightness shift, and unsupported target names using four portable inputs. The detector retains the red-moon identity across the three image perturbations, with confidence decreasing from 0.902 to 0.632 under occlusion. Unsupported names are safely rejected instead of being mapped to known objects. These input-level results define the perturbations for the next matched closed-loop evaluation.

## VI. DISCUSSION

### A. Main Finding

The central result is an attribution result, not an aggregate score. The raw video-prediction chain remains at 0/16, yet final-action state feedback replaces 23/23 planner actions and stabilizes every grounded run. More decisively, the same semantic controller and seeds 45–50 produce 6/6 successes with oracle state and 0/6 with detector/tracker feedback, with a paired visual-minus-oracle distance gap of  $0.2548 \pm 0.1453$ . Together, these comparisons identify feedback estimation and executed-action authority as the dominant tested determinants of closed-loop outcome.

The remaining experiments show that the higher-level mechanisms are independently functional. Strict no-cache language grounding reaches 9/9 target matches and joint controls, target-image grounding reaches 8/9, and forced re-query repairs 6/6 injected semantic faults with 5/6 true-target completions versus 0/6 controls. Natural re-query also fires in 3/3 enabled runs, although already-successful matched controls leave no distance margin for improvement. The execution-authority audit

TABLE X

REPEATED TARGET-IMAGE END-TO-END MATRIX. FINAL DISTANCE IS MEAN  $\pm$  SAMPLE SD OVER ALL RUNS IN THE CASE. ACTIVE TRIALS EXCLUDE RUNS ALREADY WITHIN THE 0.08 THRESHOLD BEFORE THE FIRST ACTION; JOINT ACTIVE SUCCESS REQUIRES CORRECT SEMANTIC SELECTION AND ACTIVE CONTROL SUCCESS.

| Target-image case | Runs | VLM semantic match | Pre-satisfied | Active joint success | Final distance      | Interpretation                                        |
|-------------------|------|--------------------|---------------|----------------------|---------------------|-------------------------------------------------------|
| Red moon          | 3    | 3/3                | 0             | 3/3                  | $0.0587 \pm 0.0233$ | All three trials succeed                              |
| Blue cube         | 3    | 3/3                | 0             | 3/3                  | $0.0623 \pm 0.0249$ | All three trials succeed                              |
| Yellow pentagon   | 3    | 2/3                | 0             | 2/3                  | $0.1640 \pm 0.1512$ | Seed-42 semantic error; final distance 0.3386         |
| All cases         | 9    | 8/9                | 0             | 8/9                  | $0.0950 \pm 0.0932$ | Three targets $\times$ three seeds; one semantic miss |

TABLE XI  
FACTOR-ISOLATING EVIDENCE USED TO LOCALIZE THE DOMINANT TESTED BOTTLENECK.

| Factor              | Evidence and implication                                                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw planning        | The fresh sweep is 0/16 and the recorded parameter configurations are 0/8; the tested planning variants do not reach the 0.08 completion criterion.         |
| Execution authority | Grounded runs succeed after state feedback replaces 23/23 raw actions in every run, directly identifying the feedback law as the acting controller.         |
| Feedback source     | With the semantic controller and seeds fixed, oracle state yields 6/6 and visual estimates yield 0/6; the paired distance gap is $0.2548 \pm 0.1453$ .      |
| Semantic grounding  | No-cache language trials match 9/9 targets, while target-image trials match 8/9; the retained yellow-pentagon error identifies the remaining semantic miss. |

TABLE XII  
ROBUSTNESS-EDGE AUDIT FOR TARGET DETECTION AND UNKNOWN-TARGET HANDLING.

| Case             | Status      | Confidence |
|------------------|-------------|------------|
| Original         | ok          | 0.902      |
| Occluded target  | ok          | 0.632      |
| Viewpoint proxy  | ok          | 0.902      |
| Brightness shift | ok          | 0.900      |
| Purple pole      | safe reject | —          |
| Green moon       | safe reject | —          |
| Unknown target   | safe reject | —          |

therefore converts a mixed full-pipeline outcome into specific evidence about grounding, control state, action authority, and recovery.

### B. Scope of the Finding

The attribution claim is scoped to the Language-Table simulator, the listed model assets, the evaluated controller variants, and the 0.08 project criterion. Oracle and visual feedback are compared under matched seeds, but oracle-feedback success is not assigned to the raw video-prediction planner. The 18/18 semantic aggregate measures selected-target control and is complemented by independent grounding denominators of nine language and nine target-image trials. Precision is threshold-sensitive (3/18 at 0.05), and the natural-event comparison covers already-successful runs. These boundaries preserve the component-level finding while identifying the evidence needed for broader task, image, and real-camera generalization.

### C. From Attribution to a Visual Closed Loop

The next step follows directly from the measured oracle-to-vision gap. Synchronized RGB/state traces can quantify target-

centroid error, end-effector error, goal-distance MAE, visibility F1, and temporal consistency before the detector/tracker is re-evaluated against oracle feedback on matched seeds. The same controller interface supports confidence-gated updates without changing the semantic layer, preserving the attribution design.

Recovery and grounding should then be expanded together: more instructions and target images, randomized semantic faults and failure times, and matched no-query, forced-query, and natural-event conditions. Final distance, 0.05/0.08 success, recovery latency, query count, and runtime would extend the present component-level evidence into a broader closed-loop evaluation.

### D. Data and Code Availability

The public repository at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit> and the accompanying v1.0.10 artifact include the manuscript source, authoritative 2026-09-22 and 2026-09-23 per-run and aggregate CSVs, the portable robustness-edge CSV and inputs, exact method parameters, a reproducible figure script, source and runtime-asset manifests, and an automated consistency check. The artifact deliberately excludes credentials, private endpoint addresses, local absolute paths, virtual environments, raw service logs, and third-party model weights. Reported aggregates can be recomputed from the included portable files without cached VLM responses. End-to-end reruns additionally require independently obtained detector, tracker, and video-prediction checkpoints, the ‘codex’ CLI, and a separately configured isolated ‘CODEX\_HOME’ profile for the CodexCLIClient, as documented in the reproduction guide.

## VII. CONCLUSION

We introduced an execution-authority audit that separates target selection, state source, raw planning, executed action, and closed-loop completion in a VLMPC-style Language-Table system. The audit reveals a decisive feedback-source gap: under the same semantic controller and seeds, oracle state yields 6/6 successes and detector/tracker feedback yields 0/6, while grounded logs show 23/23 raw-action replacements in every run. Independent evaluations demonstrate 9/9 language-grounding matches, 8/9 target-image matches, and 6/6 semantic-fault repairs with 5/6 true-target completions versus 0/6 controls. These results locate the current bottleneck in visual state estimation and action authority. The audit provides a reusable way to assign component responsibility before aggregate manipulation results are interpreted.

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